large near-Earth asteroid dwarfs anything else the human species can get its hands on.

As first shown by the American planetary scientist Christopher Chyba and his colleagues, little asteroids or comets, a few tens of meters across, break and burn up on entering our atmosphere. They arrive comparatively often but do no significant harm. Some idea of how frequently they enter the Earth's atmosphere has been revealed by declassified Department of Defense data obtained from special satellites monitoring the Earth for clandestine nuclear explosions. There seem to have been hundreds of small worldlets (and at least one larger body) impacting in the last 20 years. They did no harm. But, we need to be very sure we can distinguish a small colliding comet or asteroid from an atmospheric nuclear explosion.

Civilization-threatening impacts require bodies several hundred meters across, or more. (A meter is about a yard; 100 meters is roughly the length of a football field.) They arrive something like once every 200,000 years. Our civilization is only about 10,000 years old, so we should have no institutional memory of the last such impact. Nor do we.

Comet Shoemaker-Levy 9, in its succession of fiery explosions on Jupiter in July 1994, reminds us that such impacts really do occur in our time—and that the impact of a body a few kilometers across can spread debris over an area as big as the Earth. It was a kind of portent.

In the very week of the Shoemaker-Levy impact, the Science and Space Committee of the U.S. House of Representatives drafted legislation that requires NASA "in coordination with the Department of Defense and the space agencies of other countries" to identify and determine the orbital characteristics of all Earth-approaching "comets and asteroids that are greater than 1 kilometer in diameter." The work is to be completed by the year 2005. Such a search program had been advocated by many planetary scientists. But it took the death throes of a comet to move it toward practical implementation.

Spread out over the waiting time, the dangers of asteroid collision do not seem very worrisome. But if a big impact happens, it would